



Faculty of Engineering and Technology
Electrical and Computer Engineering Department
ENCS4130 – Computer Network Laboratory

Experiment No. 04

Dynamic Routing 2 – Link State Routing Protocols Open Shortest Path First (OSPF)

■ Objectives

- (a) Learn how to configure and verify IP routing on Cisco routers.
- (b) Understand the principles of dynamic routing using OSPF.

■ Requirements

- (a) Cisco Packet Tracer software.
- (b) Four routers.
- (c) Six PCs.
- (d) Three switches.

■ Pre-Lab Task

Students are required to review the Shortest Path First (SPF) Algorithm described in Section 4.2 and complete “Part One” of the problem presented in Section 4.5. In addition, they must build the network topology shown in Figure 6 and configure the IP addresses according to the guidelines provided in Section 4.4.1.

4.1 Introduction

Dynamic routing uses protocols to automatically discover networks and update routing tables. These protocols are categorized into *Interior Gateway Protocols (IGPs)*, which manage routing within an Autonomous System (AS), and *Exterior Gateway Protocols (EGPs)*, which handle routing between ASes. An **AS** is a collection of interconnected IP networks and routers under the control of a single organization, presenting a unified routing policy to the Internet.

IGPs are further classified into *distance vector (DV)* and *link-state (LS)* routing protocols. LS protocols are often considered more complex than DV protocols; however, their fundamental functionality and configuration remain straightforward. The previous experiment covered DV routing protocols (RIP and EIGRP), while this experiment focuses on LS routing protocols—specifically Open Shortest Path First (OSPF) and Intermediate System to Intermediate System (IS-IS). The EGPs will be covered in a subsequent experiment. Figure 1 illustrates the classification of dynamic routing protocols.

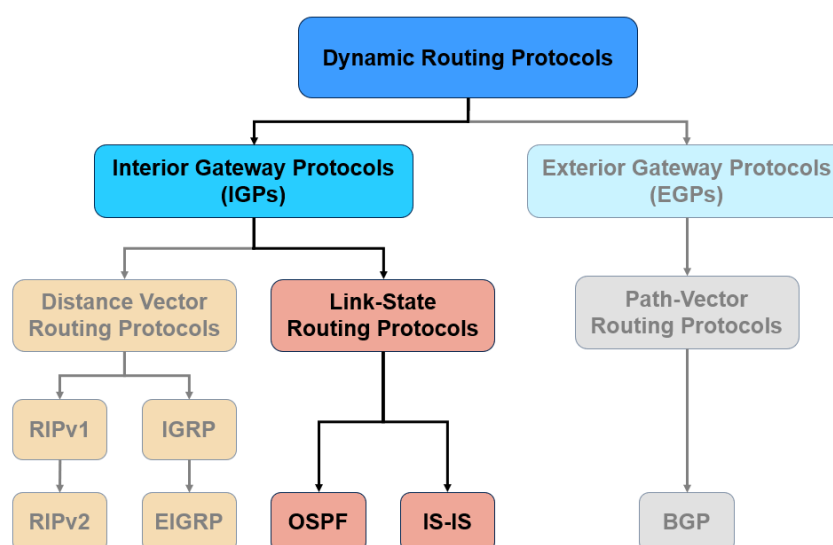


Figure 1: Dynamic routing protocols.

The term “**link**” in link-state refers to a router interface, while “**state**” describes the status of the interface and its connections to neighboring routers. LS routing protocols function as a navigation system—they *maintain a complete map* of the network. With full network visibility, routers can calculate the shortest path to any destination. This eliminates routing loops since every router has comprehensive knowledge of the topology.

Routers running an LS routing protocol follow these steps to achieve network convergence:

1. Each router detects its *active interfaces* and learns about its *directly connected networks*.
2. Routers exchange *Hello packets* to discover neighboring routers using the same LS protocol and establish adjacencies (neighbor relationships).
3. Each router creates a *Link-State Packet (LSP)*, which includes information about directly connected networks (router interfaces) and neighboring routers (e.g., neighbor ID, link type, cost).

4. Routers *flood* their LSPs to all neighbors, who store them in a *Link-State Database (LSDB)* and forward them further until all routers have the same information.
5. Using the LSDB, each router constructs a network topology map and calculates the best path to each destination using the *Shortest Path First (SPF)* algorithm.

Among LS routing protocols, OSPF is more widely used than IS-IS. One key difference is that IS-IS operates independently of the IP protocol; it exchanges routing information directly over the data link layer using the OSI Connectionless Network Service (CLNS) protocol. In contrast, OSPF runs directly over IP, which makes it easier to deploy in typical IP-based networks — which are the majority today. For this reason, the current experiment focuses primarily on OSPF.

4.2 Shortest Path First (SPF) Algorithm

All LS routing protocols use SPF algorithm, commonly known as the **Dijkstra's algorithm**, to compute the best path to each destination. The algorithm determines the optimal route by calculating the accumulated cost along each possible path from the source to the destination. Algorithm 1 presents Dijkstra's LS routing algorithm, where:

- $c(x, y)$: The direct link cost from node x to node y (∞ if they are not directly connected).
- $D(v)$: The current estimate of the least-cost path from the source to destination v .
- $p(v)$: The predecessor node along the path from the source to v .
- N' : The set of nodes whose least-cost path has been definitively determined.

Algorithm 1 Dijkstra's algorithm.

```

1: Input: Graph  $G = (V, E)$  with nodes  $V$  and edges  $E$ . Each edge  $(u, v) \in E$  has cost  $c(u, v)$ .
2: Output: Shortest path distances  $D(v)$  from source node  $u$  to all other nodes.
3: Initialization:
4:  $N' \leftarrow \{u\}$  ▷ Set of nodes with known shortest paths
5: for each node  $v \in V$  do
6:   if  $v$  is adjacent to  $u$  then
7:      $D(v) \leftarrow c(u, v)$  ▷ Initialize direct neighbor costs
8:   else
9:      $D(v) \leftarrow \infty$  ▷ Set all other distances to infinity
10:  end if
11: end for
12: while  $N' \neq V$  do ▷ Repeat until all nodes are processed
13:   Find  $w \notin N'$  such that  $D(w)$  is minimum ▷ Select closest unvisited node
14:   Add  $w$  to  $N'$ 
15:   for each neighbor  $v$  of  $w$  such that  $v \notin N'$  do
16:      $D(v) \leftarrow \min(D(v), D(w) + c(w, v))$  ▷ Update shortest path estimate
17:   end for
18: end while
19: Return  $D(v)$  for all  $v \in V$  ▷ Final shortest path distances

```

For example, consider the network topology in Figure 2, where we aim to find the best paths from node u to all other nodes. Table 1 outlines the algorithm's step-by-step execution, showing the shortest paths and accumulated costs from node u to each identified destination. Figure 3 highlights the final shortest paths after executing Dijkstra's algorithm.

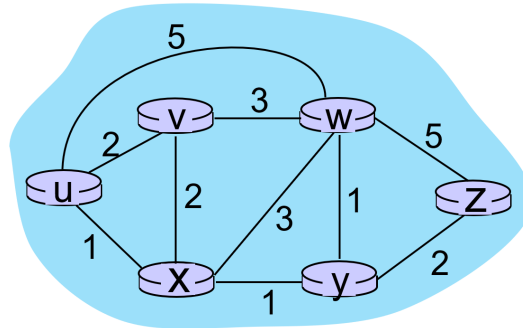


Figure 2: Sample topology (Dijkstra's algorithm).

One key observation is that this algorithm operates iteratively—after k iterations, the least-cost paths to k destinations are determined. Importantly, the shortest path is not necessarily the path with the least number of hops. For instance, consider the path to router w (and the associated w LAN). At first glance, it might appear reasonable for router u to send packets directly to router w . However, the direct path to w has a cost of 5, whereas the path via $x \rightarrow y \rightarrow w$ has a lower total cost of 4. As a result, the algorithm selects the latter as the optimal route.

Table 1: Dijkstra's algorithm table.

Step	N'	$D(v), p(v)$	$D(w), p(w)$	$D(x), p(x)$	$D(y), p(y)$	$D(z), p(z)$
0	u	2, u	5, u	1, u	∞	∞
1	u, x	2, u	4, x		2, x	∞
2	u, x, y	2, u	3, y			4, y
3	u, x, y, v		3, y			4, y
4	u, x, y, v, w					4, y
5	u, x, y, v, w, z					

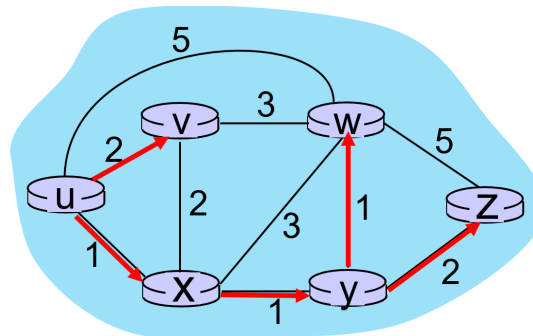


Figure 3: Sample topology (Dijkstra's algorithm) with node u SPF tree.

4.3 Open Shortest Path First (OSPF)

OSPF is a widely used link-state routing protocol known for its *scalability*, *efficiency*, and *security*. It has two defining characteristics: *openness* and *intelligent path selection*. As an ***open protocol***, OSPF is not restricted to any specific vendor, ensuring interoperability in multi-vendor network environments. It is also based on the ***SPF algorithm***, which calculates the most efficient routes for data transmission.

Designed for large and complex networks, OSPF supports *network segmentation* into multiple areas or zones, reducing routing overhead and improving organization. Unlike DV routing protocols such as RIP, which periodically exchange entire routing tables, OSPF routers share only *link-state changes*. This minimizes bandwidth consumption and enhances scalability. To further optimize routing updates, OSPF leverages *IP multicast*, ensuring efficient communication between routers.

The ability of OSPF to rapidly detect and adapt to topology changes enables faster convergence and greater network stability compared to traditional DV protocols. As a *classless routing protocol*, OSPF supports Variable Length Subnet Masks (VLSM), allowing for more efficient IP address allocation and subnetting. In addition, OSPF enhances network security by offering *packet authentication*, ensuring the integrity and authenticity of routing updates.

4.3.1 Administrative Distance (AD) and Metric

The **AD** represents the trustworthiness of routing information received by a router. It is an integer value ranging from 0 to 255, where lower values indicate more preferred routes. When multiple routes to the same destination exist, the router selects the route with the lowest AD first. Table 2 provides the default AD values for various routing protocols. For example, OSPF has an **AD** of **110**, making it more reliable than RIP (AD = 120) but less preferred than EIGRP (AD = 90) or IGRP (AD = 100). This means that if a router learns a route from both OSPF and EIGRP, it will prefer EIGRP due to its lower AD. However, despite this, OSPF remains widely used in large-scale networks due to its scalability and efficient link-state operation.

Table 2: The AD for different routing protocols.

Route Source	AD
Connected interface (directly)	0
Static route	1
EIGRP	90
IGRP	100
OSPF	110
RIP	120

When multiple routes to the same destination exist and have equal AD values, the router then evaluates *protocol-specific metrics*, commonly referred to as cost, to determine the best path. These metrics vary between routing protocols and may include hop count, bandwidth, delay, reliability, and load. In OSPF, the **metric** is based on ***bandwidth of***

the outgoing interface, with higher-bandwidth links assigned lower costs to prioritize faster paths. The cost is calculated based on a reference value using the formula:

$$\text{Cost} = \frac{\text{Reference Bandwidth}}{\text{Interface Bandwidth}} \quad (1)$$

By default, the **reference bandwidth** in OSPF is set to **100 Mbps**. For example, a 10 Mbps link has a cost of 10, while a 100 Mbps link has a cost of 1. However, links with speeds of 1 Gbps or higher also receive a cost of 1 because OSPF assigns a minimum cost value of 1, preventing further differentiation between high-speed links.

Therefore, the OSPF cost of an interface can be modified by adjusting its bandwidth. To set the bandwidth on an interface, we first access the targeted interface and then change the bandwidth using the following command:

```
Router(config-if)# bandwidth <BANDWIDTH-IN-KILOBITS>
```

The **total cost** of a route is determined by summing the cost values of each link along the path, allowing OSPF to select the most efficient route. Notably, the default cost from a *router interface to a switch* is 1.

When multiple routes have the same AD and metric, the router performs **load balancing**, distributing traffic across multiple paths. By default, Cisco routers support up to 4 *equal-cost paths*, but this can be increased to 16 equal-cost paths for OSPF.

4.3.2 Routing Hierarchy

Although LS protocols offer accurate and efficient routing, they *demand more resources* than DV protocols. LS protocols store LSPs in an LSDB and maintain an SPF tree, leading to higher memory usage. Running the SPF algorithm also demands more CPU cycles compared to DV algorithms such as Bellman-Ford. Additionally, LSP flooding can consume significant bandwidth, particularly during router startup or network instability.

To overcome these challenges, OSPF is designed with a hierarchical structure that divides an AS into multiple areas. Each area is identified by a unique *32-bit* integer Area ID, which represents a logical grouping of routers. Specifically, OSPF follows a two-level area hierarchy to organize its routing process:

- **Regular (Non-Backbone) Areas:** responsible for connecting users and resources. These areas are typically organized based on geographical or functional needs. Regular areas cannot communicate directly with each other, ensuring that non-backbone areas do not exchange routing information directly, and thus loop-free routing environment.
- **Backbone Area (known as Area 0):** serves as the core of the OSPF network, interconnecting all regular (non-backbone) areas and facilitating routing information exchange. Every non-backbone area must be connected to the backbone, either directly or via virtual links when a physical connection is unavailable. The backbone area typically does not host end-user devices.

This design introduces two types of routing: *intra-area routing*, which occurs when the source and destination are within the same area, and *inter-area routing*, which takes

place when packets need to travel between different areas, all inter-area traffic must pass through Area 0. Figure 4 shows an example of the OSPF hierarchical structure. Based on this hierarchy, we recognize different types of OSPF routers in areas:

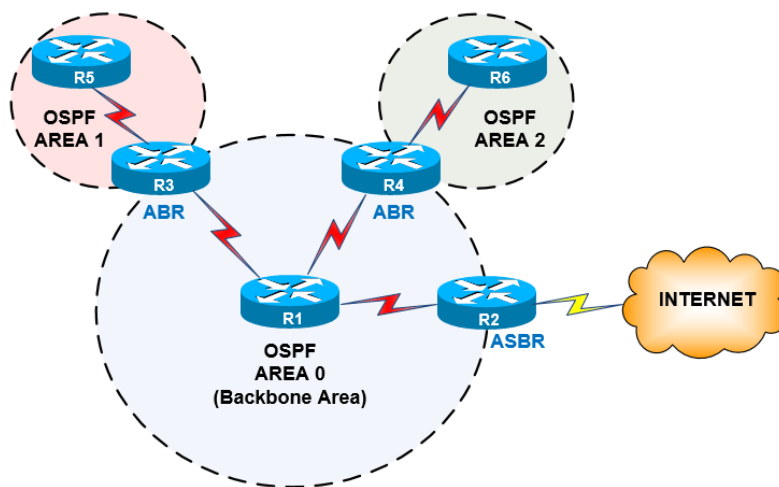


Figure 4: OSPF hierarchical structure with different types of OSPF routers.

- **Internal Routers:** have all interfaces within the same area. These routers maintain an identical LSDB to ensure consistent routing information across the area.
- **Backbone Routers:** have at least one interface in Area 0 and play a crucial role in forwarding inter-area traffic.
- **Area Border Routers (ABRs):** connect multiple OSPF areas and maintain separate LSDBs for each area they serve. An ABR collects routing information from one area, summarizes it, and advertises it to the other areas as needed. Thus, an area's internal topology remains *hidden* from other areas.
- **Autonomous System Boundary Routers (ASBRs):** connect an OSPF AS to external networks or other routing protocols such as EIGRP or BGP. These routers are responsible for redistributing external routes into the OSPF domain, ensuring seamless communication between OSPF and other routing environments.

4.3.3 Enabling OSPF

Like other routing protocols, enabling OSPF involves creating an OSPF routing process, specifying the range of IP addresses to be included in the process, and assigning appropriate Area IDs to those addresses. This configuration ensures proper routing and communication among OSPF-enabled routers.

To initiate the OSPF routing process, use the following command:

```
Router(config)# router ospf <PROCESS-ID>
Router(config-router)#
```

This command starts the OSPF process with the specified `PROCESS-ID`, a 16-bit number. Although the process ID does not need to match across routers, maintaining consistency is recommended to simplify network administration and troubleshooting.

Once the OSPF process is enabled, networks are added using:

```
Router(config-router)# network <NETWORK-ID> <WILDCARD-MASK>  
                        area <AREA-ID>
```

This command specifies the interfaces on which OSPF will operate and assigns them to an OSPF area.

4.3.4 Neighbor Relationships

Hello messages are sent on chosen interfaces once every 10 seconds on broadcast/point-to-point networks. These messages contain various pieces of information: (i) Router ID, (ii) Hello and Dead Timers, (iii) Network Mask, (iv) Area ID, (v) Neighbors, (vi) Router Priority, (vii) Designated Router (DR) / Backup Designated Router (BDR) IP Addresses, and (viii) Authentication Password. To establish an OSPF neighbor relationship, the following parameters must match between the routers:

- **Hello and Dead Timers:** These timers control how often Hello packets are sent and how long a router will wait before declaring a neighbor down.
- **Network Mask:** This parameter defines the subnet mask used for the interface. If the network mask differs, the routers will consider themselves to be on different subnets, preventing a neighbor relationship from forming.
- **Area ID:** The OSPF area identifier must match to ensure that the routers are part of the same OSPF area.

4.3.5 Loopback Interface and Router ID

A **loopback interface** is a virtual, software-based interface that emulates a physical one. Unlike physical interfaces, it is not susceptible to hardware failures and remains ‘up’ as long as it is configured. Similar to physical interfaces, a loopback interface can be assigned an IP address, making it useful for remote access via Telnet or SSH. It is widely used for routing protocols, network reachability, and testing. For example, OSPF relies on loopback interfaces to establish a stable and consistent Router ID, ensuring reliable routing operations. To configure a loopback interface, use the following commands:

```
Router(config)# interface loopback <LOOPBACK-ID>  
Router(config-if)# ip address <IP-ADDRESS> <SUBNET-MASK>
```

The loopback interface ID ranges from 0 to 10413.

The **OSPF Router ID** uniquely identifies a router within an OSPF network and must be unique. OSPF maintains network topology in its LSDB, and duplicate Router IDs can cause reachability issues. The OSPF Router ID is selected according to the following criteria, listed in order of precedence

1. Manually configured Router ID.
2. Highest IP address on a loopback interface.
3. Highest IP address on a non-loopback interface.

By default, OSPF assigns the highest IP address of any physical interface at startup unless a loopback interface exists, in which case it selects the highest IP among loopback interfaces. To manually configure the Router ID and override automatic selection, use:

```
Router(config)# router ospf <PROCESS-ID>
Router(config-router)# router-id <A.B.C.D>
```

4.3.6 Route Summarization

Route summarization is the process of consolidating multiple specific routes into a *single summary route* with an appropriate subnet mask. This optimization offers several benefits: (i) *Reduces* the size of routing update packets, (ii) *Minimizes* the routing table, improving efficiency, and (iii) *Enhances* network scalability and convergence speed.

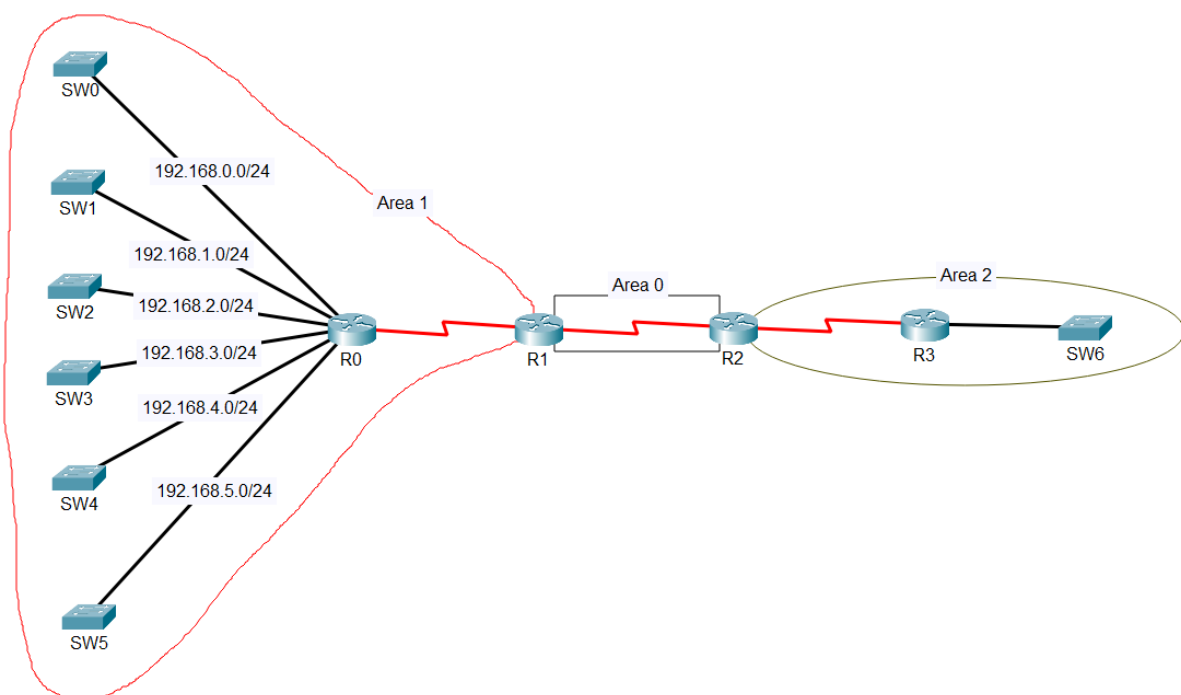


Figure 5: Route summarization (Area 1).

As illustrated in Figure 5, router R1 (an ABR) initially holds six specific routes (i.e., 192.168.0.0/24, 192.168.1.0/24, ..., 192.168.5.0/24). To optimize routing, these six networks can be summarized on the ABR into a single summary route using:

```
Router(config)# router ospf <PROCESS-ID>
Router(config-router)# area <AREA-ID> range <SUMMARY-ADDRESS>
<SUBNET-MASK>
```

To determine the appropriate summary route, follow these steps:

1. Convert the IP addresses to binary format.
2. Identify the bit position where the common pattern ends.
3. Use the first IP address in the block as the summarized route and apply the subnet mask accordingly.

Table 3: Advertised networks.

Prefix's	Binary Splitting
192.168.0.0	11000000 10101000 00000 000 00000000
192.168.1.0	11000000 10101000 00000 001 00000000
192.168.2.0	11000000 10101000 00000 010 00000000
192.168.3.0	11000000 10101000 00000 011 00000000
192.168.4.0	11000000 10101000 00000 100 00000000
192.168.5.0	11000000 10101000 00000 101 00000000

From Table 3, we can see that the first **21** bits are common across all six networks. Therefore, they can be summarized with subnet mask **/21** (i.e., 255.255.248.0), producing the summary route **192.168.0.0/21**. To configure summarization on **R1**:

```
Router(config)# router ospf 1
Router(config-router)# area 1 range 192.168.0.0 255.255.248.0
```

Although the summary route also covers networks **192.168.6.0/24** and **192.168.7.0/24**, this does not cause routing issues. For instance, if network **192.168.6.0/24** is later added to **R3** and advertised, routing will still function correctly because routers forward packets based on *longest prefix matching*.

- If a specific /24 route (e.g., 192.168.6.0/24) exists in the routing table, the router will always prefer this more specific route over the summarized /21 route. Thus, traffic destined for 192.168.6.0/24 will take the correct path without interfering with the summarized range.
- If no matching /24 route exists (e.g., 192.168.2.0/24), the router will fall back to the summarized /21 route.

4.4 Procedure

In this lab, we will connect four routers and several PCs on different networks, including loopback networks. This will require configuring routing protocols between the routers. We will use OSPF as the dynamic routing protocol.

4.4.1 Building the Topology

Construct the network topology shown in Figure 6, which consists of the following devices:

- Routers: Use **Router-PT**.
- Switches: Use **Switch-PT**.
- PCs: Use **PC-PT**.
- Connections: Use the “**Automatically Choose Connection Type**” option to link PCs, switches, and routers.

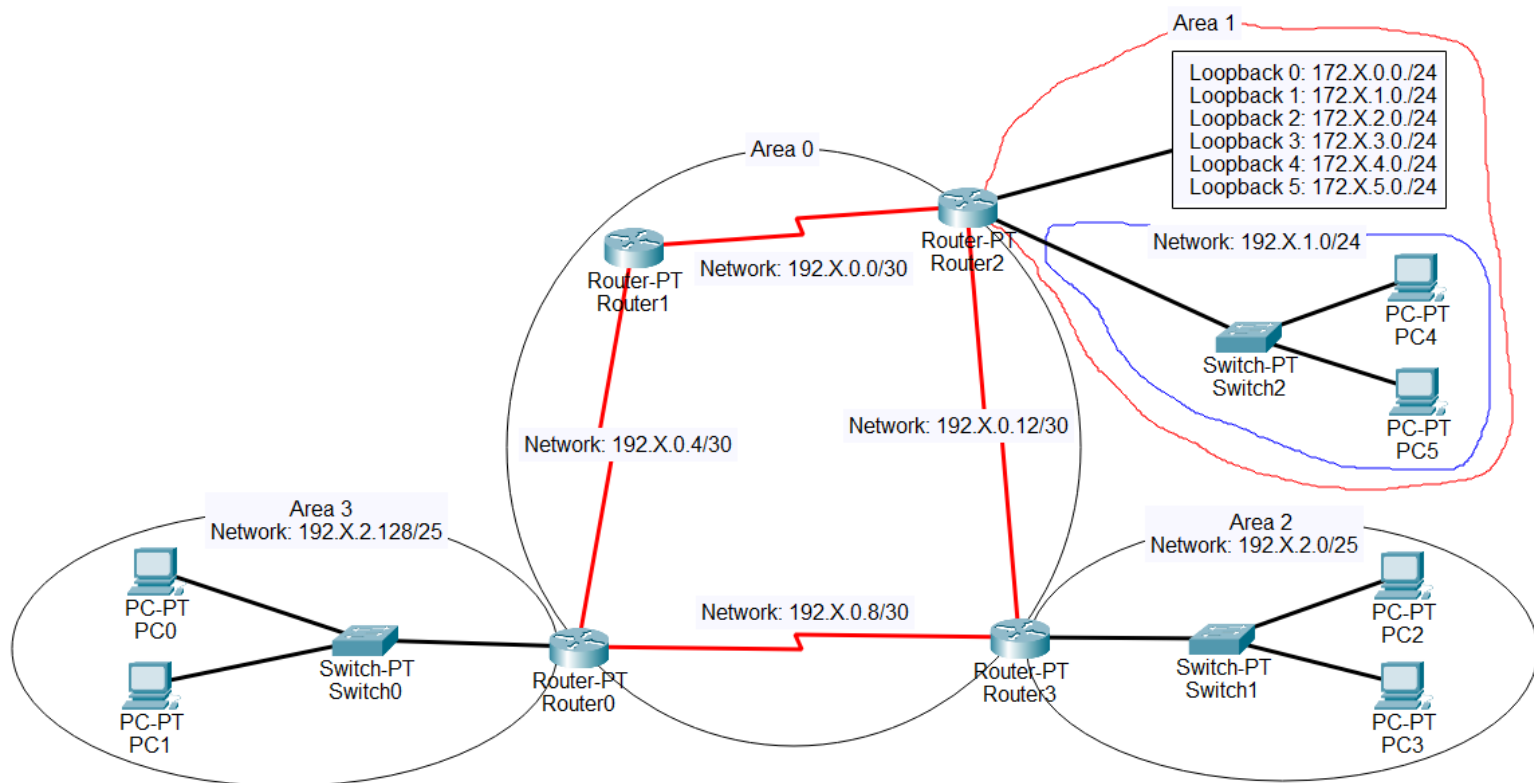


Figure 6: Dynamic routing topology.

Assign appropriate IP addresses to all PCs and routers, ensuring that each host receives a unique address within the valid host range of its respective network. The value of **X** in the network address corresponds to the *last two digits* of your student ID. For example, if the network address is **192.X.2.128/25** and your student ID is **1230105**, then **X = 05**, resulting in the network address **192.5.2.128/25**.

It is recommended to assign the first available host address (e.g., *192.X.2.129*) to router interfaces as gateways. End devices should use addresses from the remaining usable range (e.g., *PC0: 192.X.2.130*, *PC1: 192.X.2.131*, etc.) for proper network configuration.

4.4.2 Creating and Assigning an IP Address to a Loopback Interface

To configure a loopback interface, you first need to create the interface and then assign it an IP address. For example, to assign the IP address **172.X.0.1** (the first address in the **172.X.0.0/24** network) to **Loopback0**, use the following commands:

```
Router(config)# interface loopback 0
Router(config-if)# ip address 172.X.0.1 255.255.255.0
```

Now, proceed with configuring *all required* loopback interfaces.

4.4.3 Configuring OSPF Routing

The OSPF configuration requires a *Process ID*, which can be any number between 1 and 65,535. The following example shows how to enable OSPF on **Router 3** using **Process ID 1** and configure the networks accordingly:

```
Router(config)# router ospf 1
Router(config-router)# network 192.X.0.8 0.0.0.3 area 0
Router(config-router)# network 192.X.0.12 0.0.0.3 area 0
Router(config-router)# network 192.X.2.0 0.0.0.127 area 2
```

Now, configure OSPF on **Routers 0, 1, and 2** in the same manner.

Note:

To add loopback networks to OSPF on **Router 2**, each network can be configured individually using the following commands:

```
Router(config)# router ospf 1
Router(config-router)# network 172.X.0.0 0.0.0.255 area 1
Router(config-router)# network 172.X.1.0 0.0.0.255 area 1
... ..
Router(config-router)# network 172.X.4.0 0.0.0.255 area 1
Router(config-router)# network 172.X.5.0 0.0.0.255 area 1
```

This configuration can be simplified by summarizing networks:

- The first four networks (**172.X.0.0/24** to **172.X.3.0/24**) can be combined into a single **22 subnet (172.X.0.0/22)**.
- The last two networks (**172.X.4.0/24** and **172.X.5.0/24**) can be summarized into **172.X.4.0/23**.

Using these summaries, the optimized OSPF configuration becomes:

```
Router(config)# router ospf 1
Router(config-router)# network 172.X.0.0 0.0.3.255 area 1
Router(config-router)# network 172.X.4.0 0.0.1.255 area 1
```

4.4.4 Loopback Route Summarization

After configuring OSPF, **Routers 0, 1, and 3** learn about the loopback networks created on **Router 2**. These networks are installed in their routing tables as shown in Figure 7.

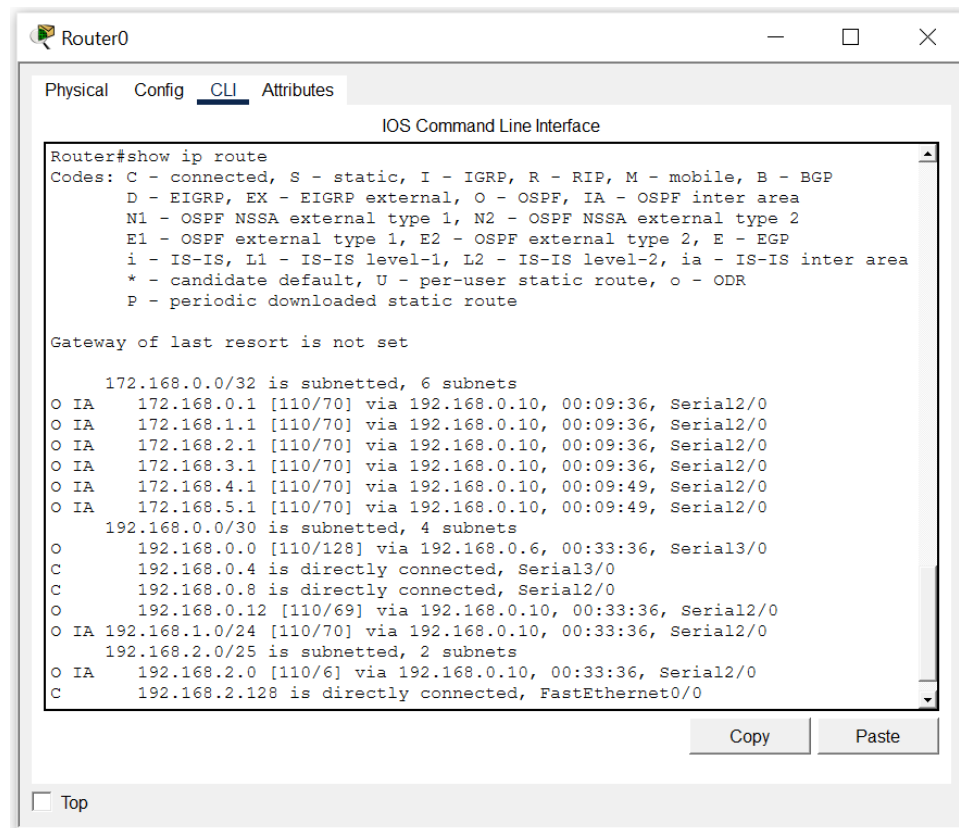


Figure 7: Routing table (Router 0) before route summarization.

To optimize routing, we can summarize the six loopback networks on the ABR into a single summary route under the OSPF process. We can see that the first **21** bits are common across all six networks. This means we can summarize them using the subnet mask **/21** (i.e., 255.255.248.0), resulting in the summarized route **172.X.0.0/21**. To configure **Router 2** (the **ABR**) for summarization, use the following commands:

```

Router(config)# router ospf 1
Router(config-router)# area 1 range 172.X.0.0 255.255.248.0
  
```

After enabling summarization on **Router 2**, **Routers 0, 1**, and **3** will receive one summarized LSP (172.X.0.0/21) instead of six individual LSPs (one for each loopback network). This reduces OSPF processing overhead and simplifies route management. In addition, by summarizing networks, we reduce the number of OSPF entries from six to one, as shown in Figure 8, improving CPU efficiency and simplifying routing.

4.4.5 Troubleshooting and Verifying Topology Configurations

Once OSPF is configured, it is crucial to verify the routing setup to ensure proper communication between routers. The following commands help inspect the OSPF configuration and monitor its behavior:

- Display the routing table:

```

Router# show ip route
  
```

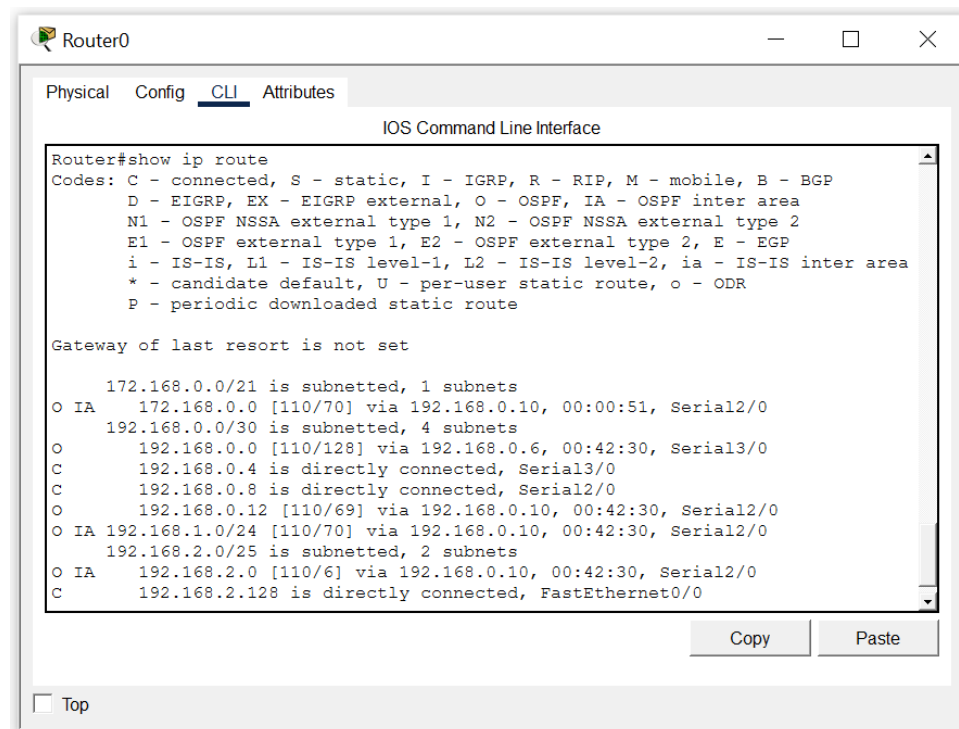


Figure 8: Routing table (Router 0) after route summarization.

This command displays the router's routing table, showing learned routes, administrative distances, next-hop addresses, and directly connected networks. It helps verify how the router forwards packets.

- **Show OSPF neighbor relationships:**

```
Router# show ip ospf neighbor
```

This command provides details about OSPF neighbors, including their router IDs, priority, state, and interface through which they are connected. It helps confirm that OSPF adjacency has been established between routers.

- **Show OSPF-enabled interfaces:**

```
Router# show ip ospf interface
```

This command displays OSPF-related interface information, such as area assignments, hello/dead intervals, cost metrics, and DR or BDR status. It ensures interfaces are correctly participating in OSPF.

- **List active routing protocols and their settings:**

```
Router# show ip protocols
```

This command shows the routing protocols running on the router, including their administrative distances, networks being advertised, and neighbor relationships. It helps verify that OSPF is active and correctly configured.

Carefully examine the output of each command and provide a clear explanation of what it indicates. Discuss your observations with your *instructor or teaching assistant (TA)*, highlighting key details.

4.4.6 Changing the Cost

Use **Cisco Packet Tracer's Simulation Mode** to analyze the path that packets take when sent from **PC0** to **PC4**. Send several packets (e.g., 4) and observe the routes they follow. Discuss your findings with your *instructor or TA*.

To control the path that packets take, you can modify the OSPF cost of specific interfaces by adjusting their bandwidth. For example, to make OSPF prefer the path **Router 0** → **Router 1** → **Router 2**, reduce the cost of one or both of the following interfaces:

- **Interface Se2/0 on Router 0**: Serial interface connecting Router 0 to Router 1.
- **Interface Se3/0 on Router 1**: Serial interface connecting Router 1 to Router 2.

The OSPF cost of an interface is calculated using the formula:

$$\text{Cost} = \frac{\text{Reference Bandwidth}}{\text{Interface Bandwidth}}$$

By default, the OSPF reference bandwidth is *100 Mbps*, and the cost on each of these interfaces is *64*. To reduce the cost to *5*, solve for the interface bandwidth:

$$5 = \frac{100}{\text{Interface Bandwidth}} \implies \text{Interface Bandwidth} = 20 \text{ Mbps} = 20000 \text{ Kbps}$$

To set the bandwidth on interface Se2/0 of **Router 0** to 20 Mbps, use:

```
Router(config)# interface Se2/0
Router(config-if)# bandwidth 20000
```

After this modification, **Router0**'s routing table will update to reflect the new cost.

Finally, use **Simulation Mode** again to confirm that packets from **PC0** to **PC4** now follow the intended path. Send several packets (e.g., 4) and observe their routes carefully. Check the routing table on **Router 0** to ensure that the updated OSPF costs are correctly applied and that the traffic follows the desired path. Discuss your results and any observations with your *instructor or TA*.

Note:

The link from the router interface to the switch has a cost of 1.

4.4.7 Important Questions

- Why are **loopback interfaces** needed in a router?
- What is the **OSPF router ID**, and why is it important?

- Manually configure the router IDs for **Router 1**, **Router 2**, and **Router 3** as **1.1.1.1**, **2.2.2.2**, and **3.3.3.3**, respectively. Verify the configuration to ensure that the changes take effect.

4.5 Problem

Given the topology shown in Figure 9, answer the following questions.

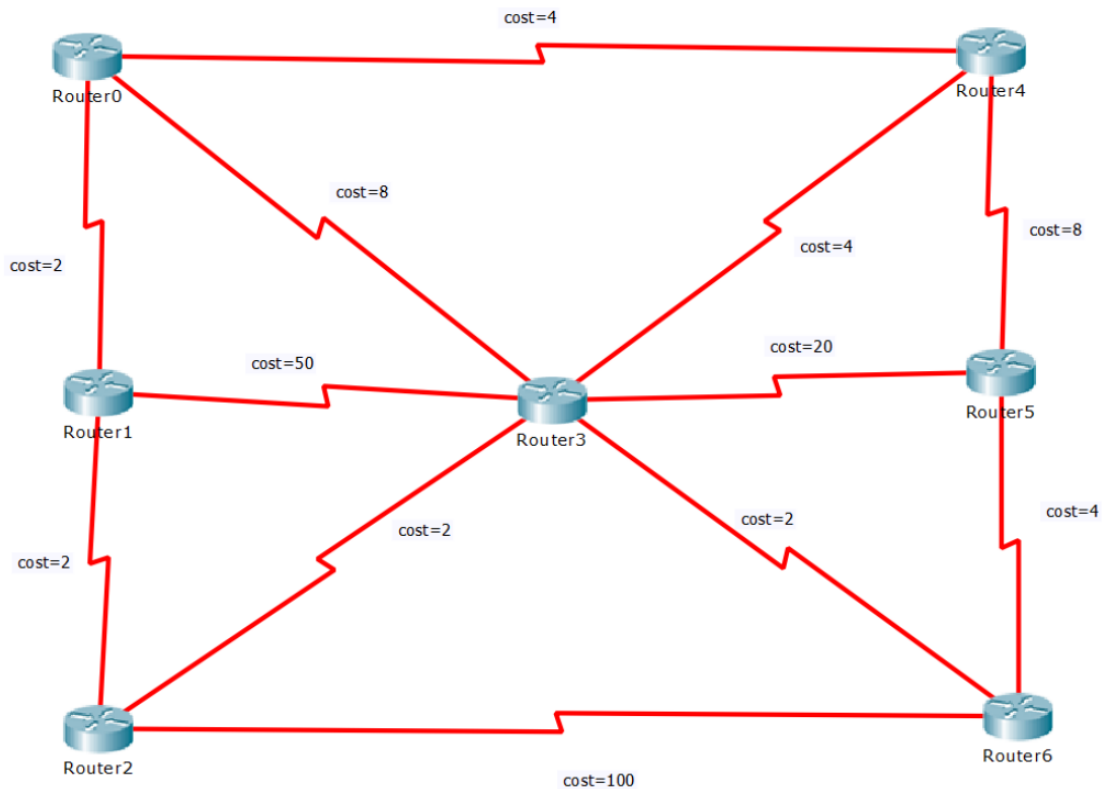


Figure 9: Problem topology.

4.5.1 Part One

1. Determine the shortest path from **Router0** to all other routers using Dijkstra's algorithm. Show all steps clearly.
2. What is the cost of the shortest path from **Router0** to **Router6**?

4.5.2 Part Two

Using Cisco Packet Tracer, build and configure the topology shown above according to the following requirements:

1. Use the Class C IP address **192.A.B.0** to address the topology, and subnet it to create networks with two usable hosts each. Here, **A** and **B** represent the last four digits of your university ID. For example, if your ID is 1230105, then $A = 01 \rightarrow 1$ and $B = 05 \rightarrow 5$.
2. Enable OSPF routing on all routers, ensuring that all belong to Area 0 (the backbone area).

3. Configure **Router6** with a loopback IP address **7.7.7.7/24** and advertise this network in the OSPF process.
4. Set the bandwidth values on all links according to the costs specified in the network topology.
5. When a packet is sent from **Router0** to **Router6** (loopback **7.7.7.7**), use the **tracert** command to identify the routers it traverses.
6. On **Router0**, use the **show ip route** command to determine the OSPF cost (metric) to reach **Router6** (loopback **7.7.7.7**). Explain how this cost is calculated.
7. Determine the Router ID of **Router0** and **Router6**, and verify your findings.

4.6 ToDo

This section will be provided by the instructor.